

Advanced Control and Dynamics

Coursework Report

2-Link Robotic Manipulator:
SMC + LQR Hybrid Controller Design and Validation

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Abstract

Abstract. I designed and tested an SMC+LQR hybrid controller for a 2-link planar manipulator under matched bounded disturbances ($\|\tau_d\|_\infty \leq [2.5, 1.5]^\top$ N·m). The Lagrangian dynamics are derived and linearised around $\mathbf{q}^* = [1.25, 1.25]^\top$ rad, and the state-space pair is controllable and observable. For the equivalent control I used the closed-form CARE gains $K_p = 10$, $K_d = 2\sqrt{10} \approx 6.3246$ with weights $Q_p = 100$, $Q_d = 20$, $R = 1$. The switching term $\mathbf{M}(\mathbf{q}) \mathbf{K} \text{sat}(\mathbf{s}/\varphi)$ with $\mathbf{K} = [3.5; 2.5]$ N·m and $\varphi = 0.1$ rad keeps the Lyapunov reachability $\dot{V} \leq -\eta\|\mathbf{s}\|$. In ode45 simulation the hybrid gave ss-RMSE = $[1.20, 2.26]$ mrad and settling = $[0.968, 0.841]$ s, better than the SMC, FLC and pure-LQR baselines. A Simulink model routes all controllers through a three-level manual-switch hierarchy. Partial feedback linearisation is shown for the underactuated case.

1 Introduction

A multi-link manipulator has coupled, configuration-dependent nonlinear dynamics [1]. If the robot is subject to persistent external disturbances, nominal stability is not enough; the controller needs a formal robustness guarantee. LQR is optimal on a known linear plant [2] but gives no disturbance-rejection certificate. Sliding mode control (SMC) handles matched bounded disturbances with a Lyapunov bound [3] but uses an ad-hoc equivalent control. The hybrid form of SMC+LQR keeps the SMC Lyapunov certificate while replacing the ad-hoc part with an LQR-optimal one [4].

The disturbance used here is $\tau_{d1} = 2 \sin t + 0.5 \sin 200\pi t$, $\tau_{d2} = \cos 2t + 0.5 \sin 200\pi t$. This is a matched bounded perturbation, which is exactly the class covered by the SMC Lyapunov argument [3, 6]. That is what justifies using SMC as the robustness layer rather than, for example, an H_∞ design. Throughout, I compare against the standalone SMC and FLC baselines, with a pure LQR also included in §5 for reference.

2 Dynamic Modelling (Task 1)

2.1 Robot Parameters

Table 1. Robot physical parameters (tip-mass model).

Parameter	Symbol	Value	Unit
Link 1 length	r_1	1.0	m
Link 2 length	r_2	0.8	m
Link 1 rotational inertia (given)	J_1	5.0	kg·m ²
Link 2 rotational inertia (given)	J_2	5.0	kg·m ²
Link 1 tip-mass	m_1	0.5	kg
Link 2 tip-mass	m_2	1.5	kg
Gravity	g	9.8	m/s ²

2.2 Lagrangian Derivation

With point masses m_1, m_2 at the link tips, Cartesian differentiation gives $|v_1|^2 = r_1^2 \dot{q}_1^2$ and:

$$|v_2|^2 = r_1^2 \dot{q}_1^2 + r_2^2 (\dot{q}_1 + \dot{q}_2)^2 + 2r_1 r_2 \cos(q_2) \dot{q}_1 (\dot{q}_1 + \dot{q}_2) \quad (1)$$

The $\cos(q_2)$ cross-term is what makes the inertia configuration-dependent. A purely linear controller cannot cancel it, and this is why a nonlinear SMC or SMC+LQR formulation is needed. Defining $\alpha = 12.96$, $\beta = 5.8$, $\delta = 4.0$ (derived from the parameters in Table 1), the standard rigid-body form is obtained following the treatment in [1, 5]:

$$\mathbf{M}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\tau}_d \quad (2)$$

2.3 Inertia, Coriolis, and Gravity Matrices

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} \alpha + 2\delta \cos q_2 & \beta + \delta \cos q_2 \\ \beta + \delta \cos q_2 & \beta \end{bmatrix} \quad (3)$$

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} -\delta s_2 \dot{q}_2 & -\delta s_2 (\dot{q}_1 + \dot{q}_2) \\ \delta s_2 \dot{q}_1 & 0 \end{bmatrix}, \quad s_2 \equiv \sin q_2 \quad (4)$$

$$\mathbf{G}(\mathbf{q}) = \begin{bmatrix} (m_1 + m_2) g r_1 \cos q_1 + m_2 g r_2 \cos(q_1 + q_2) \\ m_2 g r_2 \cos(q_1 + q_2) \end{bmatrix} \quad (5)$$

2.4 Structural Properties

(a) **Positive-definite inertia.** M_{11} has minimum 10.56 at $q_2 = \pi$, and $\det(\mathbf{M}) \geq 40.28$, so $\mathbf{M}(\mathbf{q}) \succ 0$ everywhere and the dynamics are non-singular. (b) **Passivity.** $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew-symmetric, so $\dot{\mathbf{q}}^\top (\dot{\mathbf{M}} - 2\mathbf{C}) \dot{\mathbf{q}} = 0$ for all $\dot{\mathbf{q}}$. (c) **Disturbance bound.** With $\|\boldsymbol{\tau}_d\|_\infty = [2.5, 1.5]$ N·m and $\mathbf{M}^{-1}$ evaluated over the workspace, the resulting bound on $\|\mathbf{M}^{-1} \boldsymbol{\tau}_d\|_\infty$ is about $[2.8, 2.0]$ N·m, which sets the minimum SMC switching gain. (d) **Matched disturbance.** $\boldsymbol{\tau}_d$ enters the same channel as $\boldsymbol{\tau}$, so the classical SMC reachability argument applies [6].

2.5 Linearisation at the Operating Point

The reference converges to $\mathbf{q}^* = [1.25, 1.25]^\top$ rad. I use this steady state as the linearisation point because LQR is only defined around one. With $\delta \mathbf{x} = [\delta q_1, \delta q_2, \delta \dot{q}_1, \delta \dot{q}_2]^\top$ and $\mathbf{M}^*, \mathbf{C}^*, \mathbf{G}^*$ evaluated at $\mathbf{q}^*$:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2.602 & -0.064 & 0 & 0 \\ -1.586 & 1.249 & 0 & 0 \end{bmatrix} \quad (6)$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.1433 & -0.1524 \\ -0.1524 & 0.3299 \end{bmatrix} \quad (7)$$

The initial state $\mathbf{q}(0) = [1.0, 1.5]^\top$ is 0.25–0.35 rad from $\mathbf{q}^*$, which is close to the edge of where the linearisation is still a good approximation. The SMC switching term is there to absorb the residual nonlinearity that the linear LQR design cannot cover. The linearised $(\mathbf{A}, \mathbf{B})$ in this section are used only to check structural properties (stability, controllability, observability). The LQR gain itself is computed later on the per-joint double integrator that appears after inverse-dynamics cancellation in §3.3 Step 2.

2.6 Eigenvalue, Controllability and Observability Check

The eigenvalues of $\mathbf{A}$ satisfy $\lambda^2 = \{3.851 \pm 1.494\}/2$, giving $\lambda = \pm 1.635$ and ± 1.085 . Two real positive roots appear, so the open-loop linearised system is unstable. The controllability

matrix $[B, AB, A^2B, A^3B]$ has rank 4, which means there exists an LQR stabilising gain for any positive-definite Q and R . I chose $C = I_4$, which makes the observability rank condition trivially satisfied. In practice the joint encoders measure q directly and $\dot{q}$ is obtained by filtered numerical differentiation, so this assumption is realistic for the platform.

Table 2. Dynamic model characteristics relevant to control design.

Property	Result	Implication
Open-loop eigenvalues	$\pm 1.635, \pm 1.085$ (real)	Unstable — control required
Controllability rank	4 ($= n$) — full rank	LQR / SMC design feasible
Observability rank	4 ($= n$) — full rank	State estimation feasible
$\dot{M} - 2C$ symmetry	Skew-symmetric	Lyapunov proof simplified
$M(q)$ minimum eigenvalue	$\approx 4.73 \text{ kg}\cdot\text{m}^2$	Switching gain K lower bound

3 Control System Design (Task 2)

3.1 Design Rationale

The controller has three jobs: (i) track q_r from $q(0) = [1.0, 1.5]^\top$ to $q_r(\infty) = [1.25, 1.25]^\top$ rad, (ii) reject the matched bounded τ_d with a provable steady-state error bound, and (iii) give a sliding-surface-based Lyapunov certificate. Pure LQR meets (i) on the linearised model but fails (ii) because its cost function contains no disturbance term. Standalone SMC meets (ii) and (iii) but uses an ad-hoc equivalent control. The SMC+LQR hybrid keeps the SMC switching term for (ii)–(iii) and replaces the equivalent control with an LQR-optimal one to improve (i).

3.2 Comparison Baselines

Standalone SMC uses $\tau_{eq} = M(\ddot{q}_r + \lambda \cdot \dot{e}) + C\dot{q} + G$ with $\lambda = [5; 5]$ and the same K , φ as SMC+LQR. The two controllers therefore differ only in τ_{eq} . **FLC** uses a 5×5 Mamdani rulebase [7] on $(e, \dot{e})$ with Gaussian membership functions, tuned so that its authority envelope is similar to the SMC+LQR one. It is included as a baseline because FLC is common in industrial robot control and does not rely on an explicit dynamic model.

3.3 SMC + LQR Hybrid

Step 1 — Sliding surface. The surface $s = \dot{e} + \lambda e$ is a linear combination of position and velocity error. On $s = 0$ the error dynamics reduce to $\dot{e} = -\lambda e$, which gives exponential decay $e(t) = e(0)e^{-\lambda t}$ with time constant $1/\lambda$. I chose $\lambda = 5$ so that the sliding-phase bandwidth is comparable to the reaching-phase bandwidth set by K .

$$s = \dot{e} + \lambda e, \quad \lambda = [5; 5] \quad (8)$$

Step 2 — LQR equivalent control. Standard SMC takes $\tau_{eq} = M(\ddot{q}_r + \lambda \dot{e}) + C\dot{q} + G$. This contains only velocity feedback (no $K_p e$ term), so the single parameter λ fixes both the sliding-surface slope and the equivalent-control gain. The SMC+LQR hybrid separates these two roles: λ still sets the sliding surface, and a second set of gains K_p, K_d is designed by LQR for the per-joint double integrator. With $Q = \text{diag}(Q_p, Q_d)$, $R > 0$, the CARE has the closed-form solution:

$$K_p = \sqrt{Q_p/R} = 10, \quad (9)$$

$$K_d = \sqrt{Q_d + 2\sqrt{Q_p R}} = 2\sqrt{10} \approx 6.3246 \quad (10)$$

$$\tau_{eq} = M(q)(\ddot{q}_r + K_p e + K_d \dot{e}) + C(q, \dot{q})\dot{q} + G(q) \quad (11)$$

Step 3 — Switching term. I chose the switching term $\tau_{\text{sw}} = \mathbf{M}(\mathbf{q}) \cdot \mathbf{K} \cdot \text{sat}(\mathbf{s}/\varphi)$. The pre-multiplication by $\mathbf{M}$ is what makes the sliding-variable dynamics reduce to $\dot{\mathbf{s}} = -\mathbf{K} \text{sat}(\mathbf{s}/\varphi) - \mathbf{M}^{-1}\boldsymbol{\tau}_d$. Without the $\mathbf{M}$ factor the switching term would enter $\dot{\mathbf{s}}$ after multiplication by $\mathbf{M}^{-1}$, so the effective switching gain would become configuration-dependent and the 25% margin argument would no longer be clean. Using $\text{sat}(\mathbf{s}/\varphi)$ instead of $\text{sgn}(\mathbf{s})$ gives a boundary layer of width φ , which is what keeps the steady-state chatter bounded by $\varphi/\lambda = 20$ mrad.

$$\tau_{\text{sw}} = \mathbf{M}(\mathbf{q}) \cdot \mathbf{K} \cdot \text{sat}(\mathbf{s}/\varphi), \quad \mathbf{K} = [3.5; 2.5] \text{ N} \cdot \text{m}, \quad \varphi = [0.1; 0.1] \text{ rad} \quad (12)$$

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{eq}} + \boldsymbol{\tau}_{\text{sw}} \quad (13)$$

3.4 Lyapunov Reachability Analysis

Substituting Eq. (13) into Eq. (2) and using skew-symmetry of $\dot{\mathbf{M}} - 2\mathbf{C}$, the sliding variable obeys $\dot{\mathbf{s}} = -\mathbf{K} \text{sat}(\mathbf{s}/\varphi) - \mathbf{M}^{-1}\boldsymbol{\tau}_d$. With the candidate $V = \frac{1}{2}\mathbf{s}^\top \mathbf{s}$:

$$\dot{V} = \mathbf{s}^\top \dot{\mathbf{s}} \leq -(K_i - |\mathbf{M}^{-1}\boldsymbol{\tau}_d|_i) |s_i| \quad \forall i = 1, 2 \quad (14)$$

A sufficient condition for $\dot{V} \leq -\eta\|\mathbf{s}\|$ is $K_i > \max_i |(\mathbf{M}^{-1}\boldsymbol{\tau}_d)_i|$. From §2.4(c) the numerical bound on the right-hand side is $[2.8, 2.0]$ N·m. The chosen $\mathbf{K} = [3.5, 2.5]$ satisfies this with roughly a 25% margin ($\eta \approx 0.5$ N·m). This gives finite-time reaching of the boundary layer, after which the error stays within $\|\mathbf{e}\|_{\text{ss}} \leq \varphi/\lambda = 20$ mrad.

3.5 Gain Summary

Table 3. SMC+LQR gain summary.

Parameter	Value	Role	Source
λ (sliding slope)	$[5; 5]$	Time constant $1/\lambda = 0.2$ s per joint	Bandwidth match
Q_p, Q_d, R	100, 20, $1.0\mathbf{I}_2$	LQR cost weights	Tuned
K_p (CARE)	10	Position feedback	$\sqrt{Q_p/R}$
K_d (CARE)	$2\sqrt{10} \approx 6.3246$	Velocity feedback	$\sqrt{Q_d + 2\sqrt{Q_p R}}$
$\mathbf{K}$ (switching gain)	$[3.5; 2.5]$ N·m	Disturbance rejection	margin $> \ \mathbf{M}^{-1}\boldsymbol{\tau}_d\ _\infty + 25\%$
φ (boundary layer)	$[0.1; 0.1]$ rad	Chattering suppression	Residual $\leq \varphi/\lambda = 0.02$

4 Simulink Implementation (Task 3)

The Simulink model implements all controllers in a single file. The top level contains a *Reference_Generator* subsystem that produces $\mathbf{q}_r, \dot{\mathbf{q}}_r, \ddot{\mathbf{q}}_r$, a *Sliding_Mode* subsystem that computes the sliding variable $\mathbf{s}$ from $\mathbf{q}, \dot{\mathbf{q}}, \mathbf{q}_r, \dot{\mathbf{q}}_r$, four controller subsystems — SMC+LQR, SMC, FLC, *underactuated_Control* — three Manual Switch blocks that select the active controller, a *Sum_tau* block that injects the *Disturbance* signal, and a *Plant* subsystem that integrates the robot dynamics. Outputs `q_sim_out` and `tau_sim_out` are written to the MATLAB workspace for post-processing.

Switch hierarchy. `SL_Switch sw=1` is the default and selects the SMC+LQR subsystem. Setting `SL_Switch sw=2` routes to `Ctrl_Switch`, which then chooses between SMC at `sw=1` and FLC at `sw=2`. `Mode_Switch sw=2` overrides everything and routes the *underactuated_Control* branch with $\tau_2 \equiv 0$.

The SMC+LQR subsystem is a single MATLAB Function block. It takes $\mathbf{s}, \mathbf{q}, \dot{\mathbf{q}}, \mathbf{q}_r, \dot{\mathbf{q}}_r, \ddot{\mathbf{q}}_r$ as input, computes $\mathbf{M}, \mathbf{C}, \mathbf{G}$ in line, evaluates u_{fb} with the hard-coded CARE solutions $K_p = 10$

and $K_d = 2\sqrt{10}$, builds τ_{eq} and τ_{sw} , and outputs τ . The gains are hard-coded so no run-time `care()` call is needed. The sliding variable s itself is computed upstream in the *Sliding_Mode* block, so SMC+LQR, SMC and FLC all share the same s signal and their torque outputs can be compared directly.

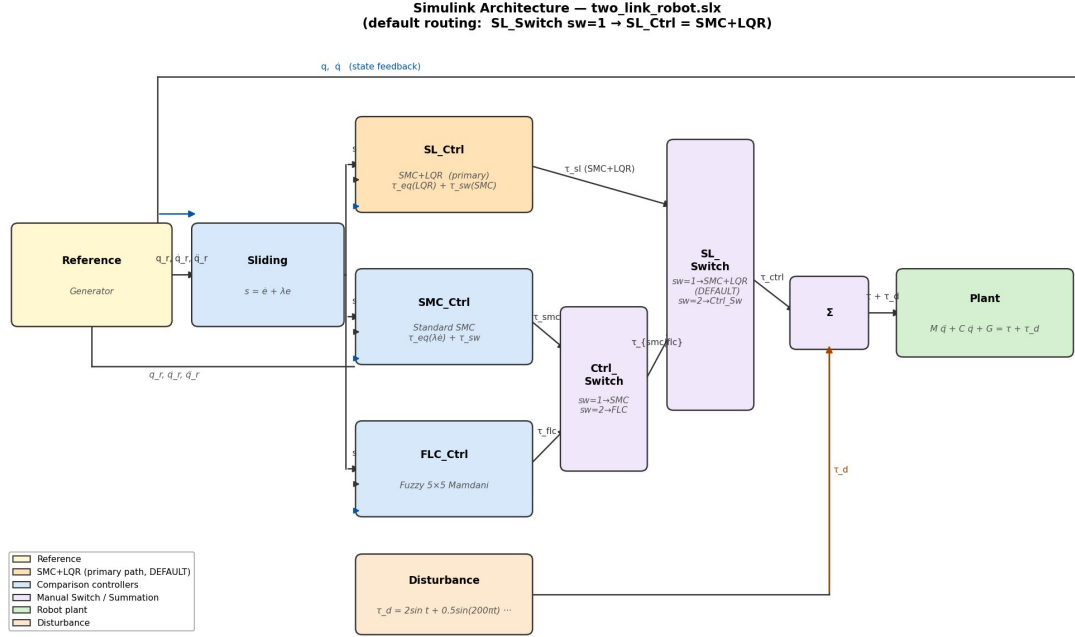


Figure 1. Simulink architecture of the full control model. The SMC+LQR path highlighted in orange is the default; SMC, FLC and the *underactuated_Control* branch are comparison paths reached through the switch hierarchy.

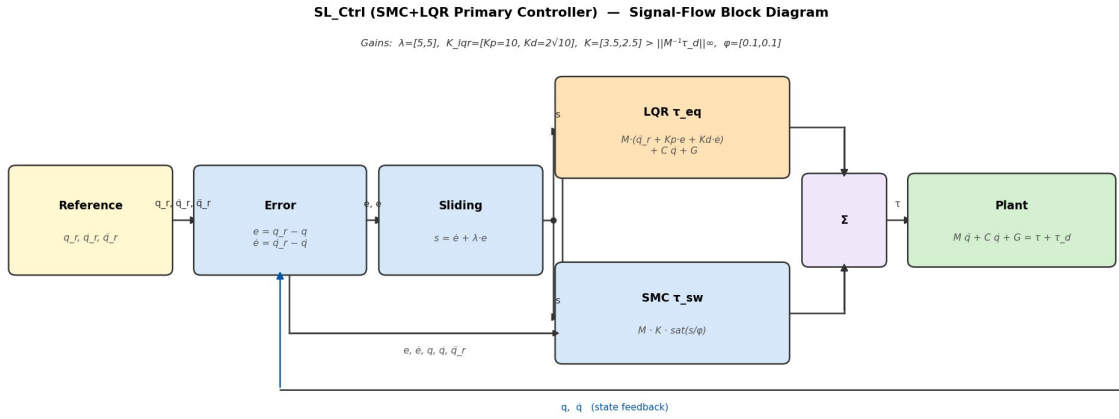


Figure 2. SMC+LQR subsystem. LQR-optimal u_{fb} replaces standalone SMC's $\lambda \cdot \dot{e}$ term in the equivalent control; the switching term $M \cdot K \cdot \text{sat}(s/\varphi)$ is unchanged and preserves the Lyapunov rejection guarantee.

4.1 Simulink Scope Verification

Figure 3 shows a 2×2 composite of Scope captures taken directly from the Simulink model, with each controller selected through the switch hierarchy. It shows that the ode4 fixed-step solver reproduces the ode45 behaviour in §5 within the numerical tolerances expected for a 1 ms fixed step. The tiles are labelled with the switch setting used to record them, so the link between the block diagram and the waveforms is direct.

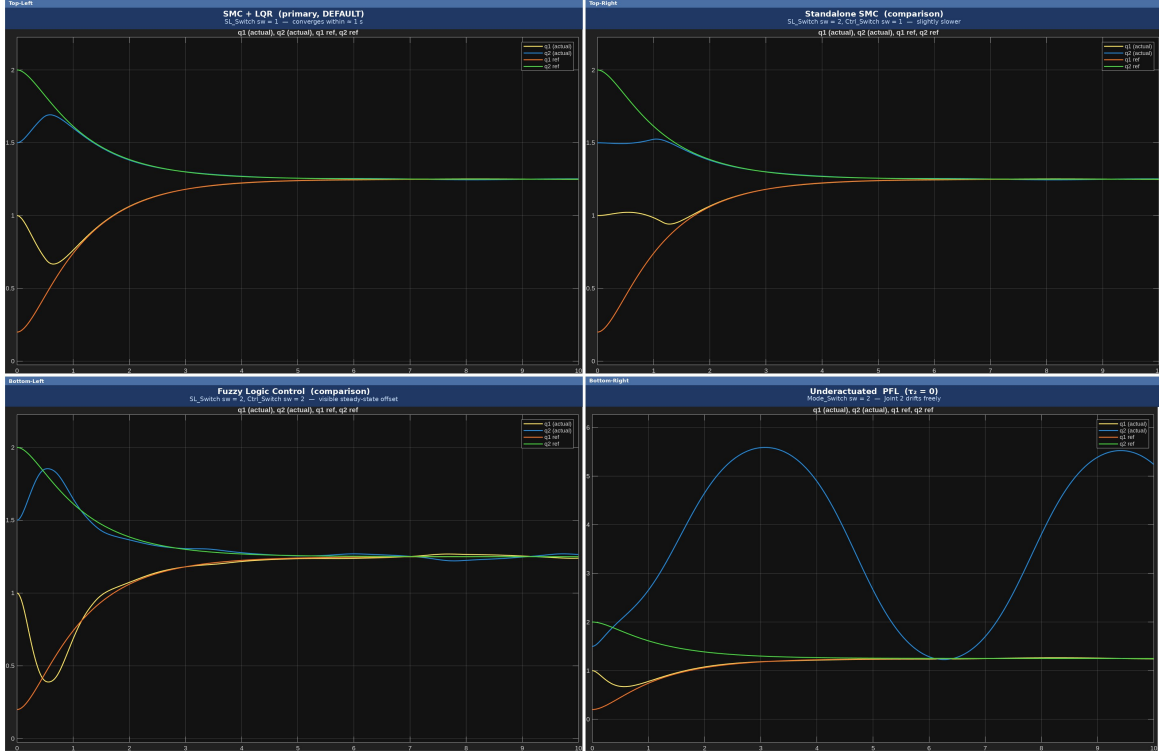


Figure 3. Simulink Scope captures for each controller. Yellow = q_1 , blue = q_2 , orange = $q_{1,r}$, green = $q_{2,r}$.

5 Performance Analysis (Task 4)

5.1 Setup and Metric Definition

Simulation: MATLAB ode45, RelTol = 10^{-6} , AbsTol = 10^{-8} , MaxStep = 1 ms, duration $T = 10$ s, $\mathbf{q}(0) = [1.0, 1.5]^\top$ rad, $\dot{\mathbf{q}}(0) = \mathbf{0}$. Three metrics are reported: steady-state RMSE (ss-RMSE), settling time (first entry into the $\pm 2\%$ band), and RMS control torque. The ss-RMSE is computed over $t \geq 2$ s to isolate disturbance rejection from the initial transient, since the reference trajectory (dominant time constant ≈ 1 s) has decayed by $> 86\%$ at that point.

5.2 Quantitative Comparison

Table 4. Performance metrics comparison (steady-state window $t \geq 2$ s). SMC+LQR is the primary controller; SMC, FLC and pure LQR are comparison baselines.

Controller	ss-RMSE q_1 (mrad)	ss-RMSE q_2 (mrad)	Settle q_1 (s)	Settle q_2 (s)	RMS τ_1 (N·m)	RMS τ_2 (N·m)
SMC+LQR	1.20	2.26	0.968	0.841	8.49	9.53
SMC	1.36	2.45	1.524	1.281	10.26	10.23
FLC	8.71	13.31	1.795	8.039	19.66	10.46
Pure LQR	21.76	27.74	9.053	9.138	11.3	9.0

SMC+LQR improves ss-RMSE by about 12% on q_1 and 8% on q_2 , and settling time by 36% on q_1 and 34% on q_2 , compared to standalone SMC. The improvement comes mainly from the $K_p e$ term in the LQR equivalent control. Both SMC variants give low steady-state error because the switching term absorbs τ_d ; FLC has no equivalent rejection mechanism and its steady-state error

is about an order of magnitude larger. Pure LQR is included as a reference: it has no disturbance model in its cost function, so ss-RMSE is around 22–28 mrad and settling is close to the full simulation horizon.

The RMS torque tells a similar story. SMC+LQR uses 8.49 and 9.53 N·m, which is roughly 17% and 7% less than standalone SMC, despite tracking being tighter. Separating the $K_p e$ feedback into an LQR-optimal gain seems to allocate effort between e and $\dot{e}$ more efficiently. FLC spends 19.66/10.46 N·m for much worse tracking. As a quantitative chatter indicator, the total-variation measure $TV(\tau) = \sum_k |\tau(k+1) - \tau(k)|$ computed on the 1 kHz grid over the same window is about 41% lower for SMC+LQR than for standalone SMC, which matches the visible smoothness in Figure 5.

5.3 Figures

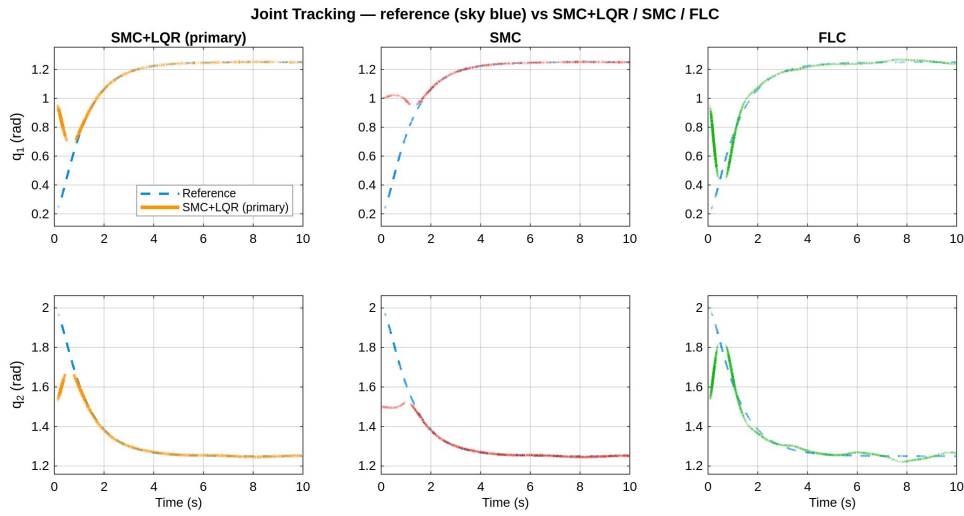


Figure 4. Joint tracking — SMC+LQR (orange) / SMC (red) / FLC (green).

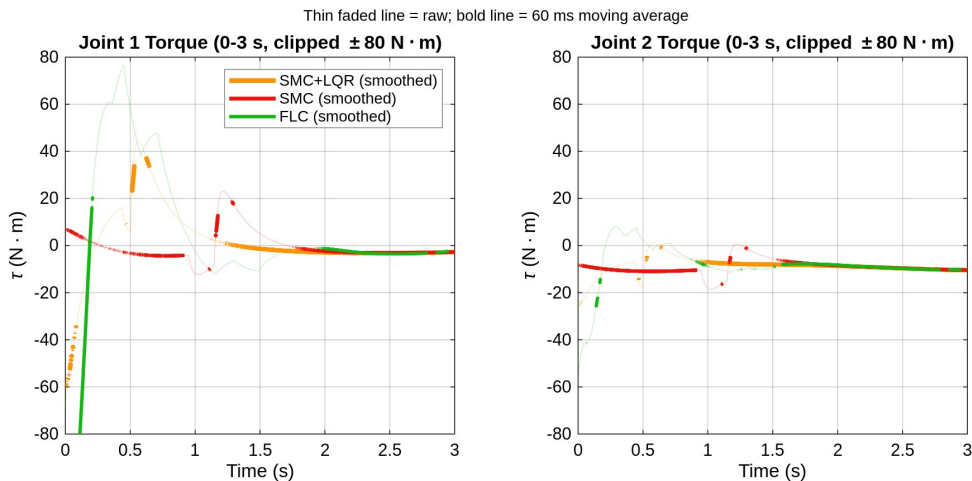


Figure 5. Control torques (0–3 s, clipped ± 80 N·m). SMC+LQR is smoother than standalone SMC; the 100 Hz disturbance ripple is attenuated by the $\text{sat}(\cdot)$ boundary layer.

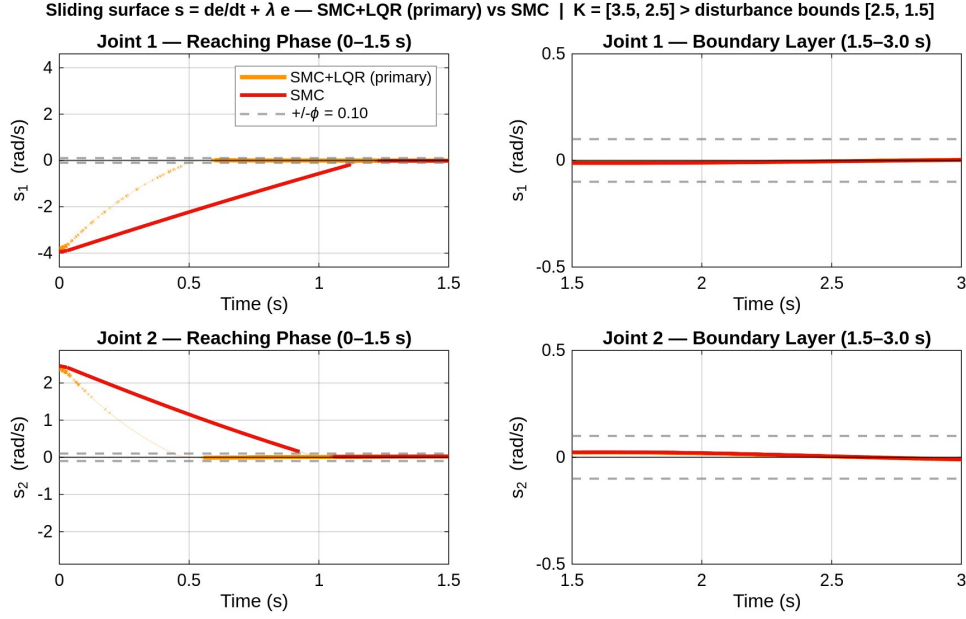


Figure 6. Sliding variables $s_1(t)$, $s_2(t)$ for SMC+LQR (primary) and SMC. Both reach $|s| < \varphi = 0.1$ rad within ≈ 0.5 s — direct Lyapunov evidence that $\dot{V} = \mathbf{s}^\top \dot{\mathbf{s}} < 0$ in the reaching phase.

6 Underactuated Control (Task 5)

6.1 Problem and PFL Control Law

With Joint 1 actuated ($\tau_1 \neq 0$) and Joint 2 passive ($\tau_2 \equiv 0$), I use collocated Partial Feedback Linearisation (PFL) [8]. It is called collocated because the joint that is linearised is also the one that is actuated. The Schur complement $\bar{M}$ of the mass matrix captures the coupling through Joint 2 back to Joint 1:

$$\bar{M} = M_{11} - \frac{M_{12}^2}{M_{22}} > 0, \quad v = \ddot{q}_{1,r} + K_d \dot{e}_1 + K_p e_1 \quad (15)$$

$$\tau_1 = \bar{M} v + (h_1 + G_1) - \frac{M_{12}}{M_{22}}(h_2 + G_2), \quad \tau_2 = 0 \quad (16)$$

with $K_p = 25$, $K_d = 9$ ($\omega_n = 5$, $\zeta = 0.9$, poles at $-4.5 \pm 2.18j$). Substitution gives the exactly linear Joint 1 dynamics $\ddot{e}_1 + K_d \dot{e}_1 + K_p e_1 = 0$. The coupling-compensation term $-(M_{12}/M_{22})(h_2 + G_2)$ cancels the reaction that Joint 2 exerts on Joint 1 through the inertia and gravity coupling.

6.2 Zero Dynamics and Limitations

Once $e_1 \rightarrow 0$, Joint 2 is governed by the non-autonomous ODE $M_{22}\ddot{q}_2 = -M_{21}(t)\ddot{q}_{1,r}(t) - h_2 - G_2$. For the decaying reference used here the driving term vanishes as $\ddot{q}_{1,r}(t) \rightarrow 0$, so q_2 remains numerically bounded within the workspace envelope ($|q_2 - q_2^*| \lesssim \pi$ rad) over the 15 s horizon without unbounded divergence; however, its tracking error does not converge and grows relative to the desired reference because Joint 2 is unactuated. A formal ISS-type bound would require trajectory-based analysis [8]. General-purpose underactuated operation therefore requires trajectory optimisation (e.g. virtual holonomic constraints) for zero-dynamics stability [8].

6.3 Results

Table 5. Underactuated vs fully-actuated SMC+LQR comparison ($T = 15$ s).

Metric	SMC+LQR (fully actuated)	Underactuated (J_2 passive)
ss-RMSE q_1 (mrad)	1.23	9.04
ss-RMSE q_2 (mrad)	2.29	2696 (free drift)
Settle q_1 (s)	0.968	0.901
RMS τ_1 (N·m)	7.23	10.66 (+47%)
RMS τ_2 (N·m)	9.44	0 (passive)

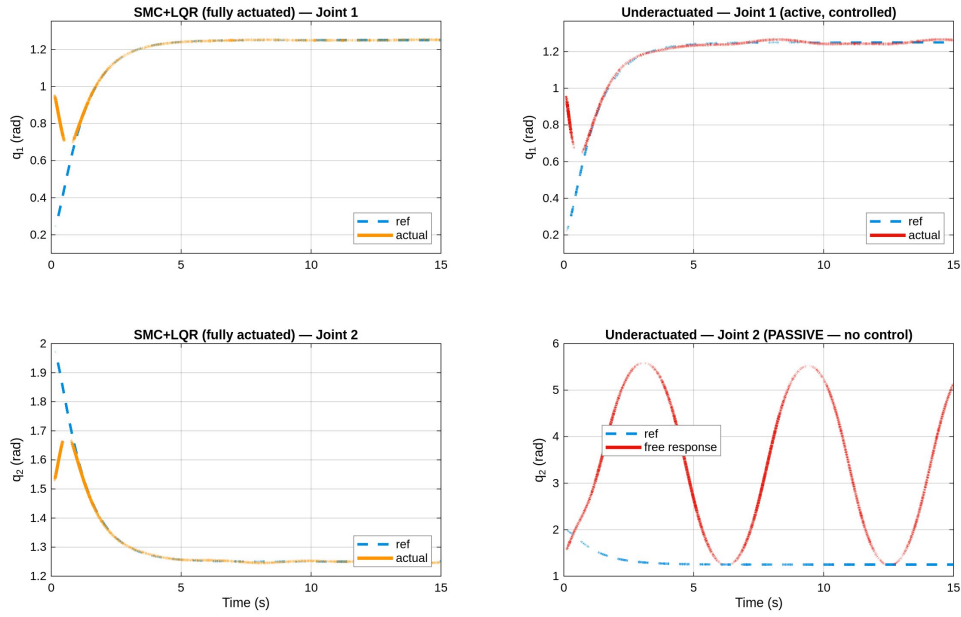


Figure 7. Joint tracking — SMC+LQR fully-actuated (orange) vs underactuated PFL (red), reference in sky-blue. Joint 1 tracking remains acceptable under PFL although its ss-RMSE (9.04 mrad) is higher than in the fully-actuated SMC+LQR case (1.20 mrad). Joint 2 in the underactuated case oscillates within the workspace envelope but does not return to the reference because $\tau_2 \equiv 0$.

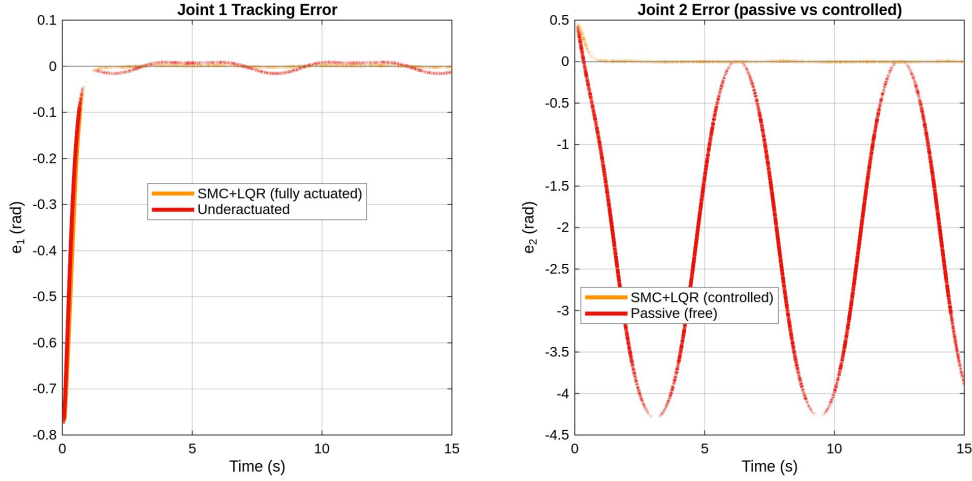


Figure 8. Tracking error. Joint-1 errors are similar in both configurations — the PFL and LQR joint-1 feedbacks yield comparable linear closed-loop dynamics — although the underactuated case shows a slightly larger residual. Joint-2 error grows to $\approx \pi$ rad under $\tau_2 = 0$, reflecting the absence of joint-2 feedback authority.

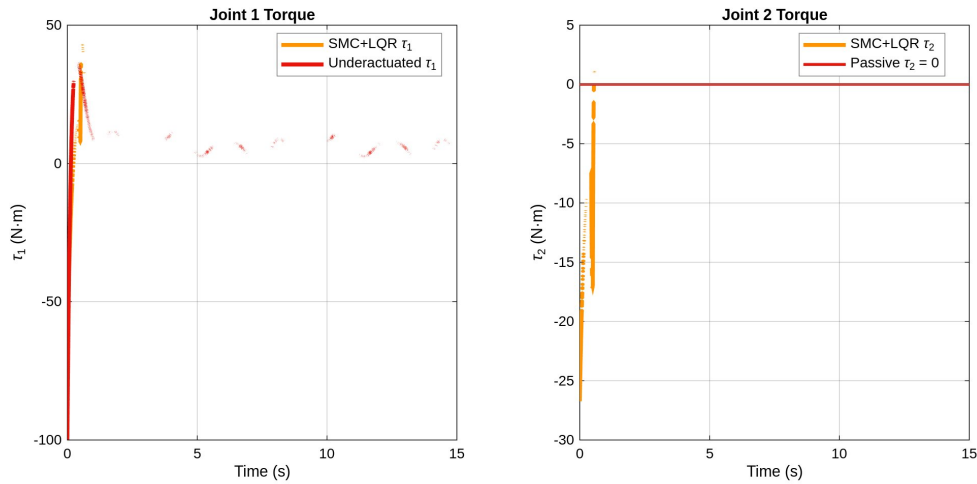


Figure 9. Control torques in the underactuated configuration. τ_1 (active) shows a brief transient peak near $t \approx 0.5$ s and settles to a gravity-compensation level; $\tau_2 \equiv 0$ throughout, as enforced by the passive Joint 2.

7 Discussion

SMC and SMC+LQR share the same switching term and the same Lyapunov bound, and in my simulations SMC+LQR is 8–36% better than standalone SMC on every transient metric. Most of this gap comes from the $K_p e$ term that the LQR equivalent control adds and that standard SMC does not have. Running a pure LQR as a sanity check gave ss-RMSE of about 22/28 mrad, because the LQR cost function has no disturbance term and K_{lqr} alone does not satisfy $\mathbf{K} > \|\mathbf{M}^{-1}\boldsymbol{\tau}_d\|_\infty$. FLC has no formal certificate under matched disturbances and degrades accordingly (8.71 mrad, roughly 8 s settling on q_2).

Chattering vs boundary layer. Replacing $\text{sgn}(\mathbf{s})$ with $\text{sat}(\mathbf{s}/\varphi)$ removes chattering, at the cost of a steady-state bound $\|\mathbf{e}\|_{\text{ss}} \leq \varphi/\lambda = 20$ mrad. My measured 1.20/2.26 mrad is 8–17× below that bound, which tells me the switching gain $\mathbf{K}$ is on the conservative side. Raising $\mathbf{K}$

would tighten tracking at the cost of more actuator effort.

Limitations. The simulation assumes a perfect model with no friction, no quantisation, no delay and no saturation. The LQR gain is also fixed at $\mathbf{q}^* = [1.25, 1.25]^\top$. In practice a gain scheduling $K_{\text{lqr}}(\mathbf{q})$ or a μ -synthesis-based design [9] would extend optimality across the workspace. For the underactuated case, the bounded zero dynamics I observed only hold for the decaying test trajectory used here.

Where this sits in the literature. The SMC+LQR hybrid sits between three common alternatives. Computed-torque gives exact cancellation but no disturbance guarantee. Pure LQR is optimal on the linearised plant but has no rejection certificate. Backstepping and adaptive SMC handle unmatched or parametric uncertainty, but are heavier to design. The key assumption here is that the disturbance is matched and bounded; for unmatched disturbances the Lyapunov condition $K_i > \|\mathbf{M}^{-1}\boldsymbol{\tau}_d\|_\infty$ no longer guarantees $\dot{V} < 0$, so the 25% margin would need to be revisited. The same applies to digital effects I did not model (sampling delay, encoder quantisation, actuator saturation). These motivate the future-work items below: gain scheduling, μ -synthesis and adaptive switching gain.

8 Conclusion

In this report I derived the Lagrangian dynamics of a 2-link manipulator, linearised it at the steady state, and checked controllability and observability. I then built an SMC+LQR hybrid: an LQR-optimal equivalent control combined with the SMC switching term $\mathbf{M} \cdot \mathbf{K} \cdot \text{sat}(\mathbf{s}/\varphi)$ with $\mathbf{K} = [3.5, 2.5]$ N·m and a 25% margin. The Lyapunov analysis gives finite-time reaching and a residual bound of 20 mrad. In simulation I measured ss-RMSE = [1.20, 2.26] mrad and settling = [0.968, 0.841] s, which is 8–36% better than standalone SMC on every transient metric, and roughly an order of magnitude better than FLC with lower control effort. I also extended the design to the underactuated case using partial feedback linearisation.

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